

TA-GraphRAG: Trust-Aware Adaptive Graph Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) systems have emerged as a promising paradigm for enhancing large language models (LLMs) with external knowledge. However, existing RAG approaches primarily focus on retrieval accuracy and generation quality, largely overlooking the *trustworthiness* of the entire pipeline. Meanwhile, GraphRAG methods leverage structured knowledge graphs for more precise retrieval but remain vulnerable to noise, adversarial inputs, and knowledge conflicts. In this paper, we propose **TA-GraphRAG** (Trust-Aware Adaptive Graph Retrieval-Augmented Generation), a novel framework that systematically integrates trustworthiness constraints across multiple dimensions into every stage of the GraphRAG pipeline. Our framework consists of six key modules: (1) an adaptive retrieval judge that decides when and what to retrieve based on confidence and trust scores; (2) a multi-granularity graph retriever with hybrid search strategies; (3) a trust-aware knowledge filter that validates retrieved evidence across multiple trust dimensions; (4) a graph-enhanced generator that leverages structural knowledge for grounded generation; (5) a three-stage hallucination detector operating before, during, and after generation; and (6) an iterative refinement loop that triggers re-retrieval when trust scores fall below thresholds. A pilot evaluation on HotpotQA (10 multi-hop questions) validates the end-to-end pipeline, with the trust filter achieving a mean trust score of 0.87 and the hallucination detector converging in an average of 1.2 iterations.

Keywords: Retrieval-Augmented Generation, Knowledge Graph, Trustworthiness, Hallucination Detection, Adaptive Retrieval

1 Introduction

Large language models (LLMs) have achieved remarkable success across a wide range of natural language processing tasks [2, 6]. However, they inherently suffer from several critical limitations: hallucination [13], outdated knowledge [7], and lack of transparent reasoning processes [17]. Retrieval-Augmented Generation (RAG) [8] has emerged as a promising solution by incorporating relevant documents from external knowledge bases to ground the generation process.

Despite the rapid advancement of RAG techniques [20, 12], several fundamental challenges remain unresolved:

1. **Retrieval Noise and Robustness.** Retrieved documents often contain irrelevant, outdated, or even adversarial information. Existing RAG systems lack systematic mechanisms to filter untrustworthy evidence before it influences generation [11].
2. **Knowledge Conflicts.** When retrieved external knowledge contradicts the parametric knowledge stored in LLMs, models often exhibit a strong confirmation bias toward their pre-trained knowledge [10], leading to unreliable outputs.

3. **Trustworthiness Blind Spots.** Current evaluation of RAG systems predominantly focuses on answer accuracy (e.g., EM, F1), neglecting critical dimensions such as fairness, transparency, and accountability [24].
4. **Flat Retrieval vs. Structured Knowledge.** Most RAG systems treat retrieved documents as flat text, ignoring the rich relational structure inherent in knowledge graphs that could enable more precise and verifiable reasoning [11].

To address these challenges, we propose **TA-GraphRAG**, a Trust-Aware Adaptive Graph Retrieval-Augmented Generation framework that systematically integrates multi-dimensional trustworthiness constraints into a GraphRAG pipeline. The central insight is that *trustworthiness cannot be retrofitted as a post-hoc filter*—it must be embedded as a first-class constraint at every stage, from the initial retrieval decision through iterative refinement. Our key contributions are:

1. We propose a **unified framework** that embeds multi-dimensional trustworthiness constraints into every stage of the GraphRAG pipeline, from retrieval decision to output verification. Inspired by the trust dimensions defined by Zhou et al. [24], we implement four filtering dimensions in Module 3: factuality (NLI-based verification), robustness (perturbation testing), fairness (bias detection), and recency (temporal weighting).
2. We design a **trust-aware adaptive retrieval mechanism** that jointly considers semantic relevance and trustworthiness scores to decide whether, what, and how to retrieve, enabling a “retrieve-evaluate-supplement” closed loop.
3. We develop a **three-stage hallucination detection module** that combines pre-generation conflict detection, during-generation contrastive decoding, and post-generation graph-path verification for structured fact-checking.
4. A pilot evaluation on HotpotQA validates the complete pipeline, demonstrating that the trust filter selectively retains high-quality evidence (mean trust score 0.87) while the hallucination detector converges efficiently (1.2 iterations on average).

2 Related Work

2.1 Retrieval-Augmented Generation

RAG was first introduced by Lewis et al. [8], combining a pre-trained seq2seq model with a dense passage retriever. Subsequent work has advanced RAG along several dimensions. Gao et al. [20] categorized RAG paradigms into Naive RAG, Advanced RAG, and Modular RAG, highlighting the trend toward more flexible and modular architectures. Zhao et al. [12] further surveyed RAG applications across diverse modalities including text, code, image, and audio, establishing a “foundation-enhancement-application” taxonomy.

Adaptive retrieval methods such as FLARE [22] and Self-RAG [1] enable LLMs to dynamically decide when to retrieve, improving efficiency and reducing unnecessary retrieval. However, these methods rely primarily on confidence thresholds and do not incorporate trustworthiness signals.

2.2 Graph Retrieval-Augmented Generation

GraphRAG leverages structured knowledge graphs to enhance RAG systems. Peng et al. [11] provided a systematic survey formalizing the GraphRAG workflow into three stages: Graph-Based Indexing (G-Indexing), Graph-Guided Retrieval (G-Retrieval), and Graph-Enhanced Generation (G-Generation). Key advantages include multi-granularity retrieval (entity, triplet,

path, subgraph levels) and the ability to capture relational structures missed by flat text retrieval.

However, GraphRAG methods face challenges including scalability to large knowledge graphs, handling dynamic knowledge updates, and—critically—lack of trustworthiness guarantees during retrieval and generation.

2.3 Trustworthiness in LLMs and RAG Systems

The trustworthiness of LLMs has been studied across multiple dimensions. Zhou et al. [24] proposed a unified framework assessing RAG trustworthiness across six dimensions: factuality, robustness, fairness, transparency, accountability, and privacy, along with the TRC Bench benchmark. Tonmoy et al. [16] surveyed over 32 hallucination mitigation techniques, categorizing them into prompt engineering and model development approaches.

Despite these advances, trustworthiness considerations have been largely absent from the design of GraphRAG systems. Our work bridges this gap by systematically integrating trust constraints into the GraphRAG pipeline.

2.4 Knowledge Conflict and Hallucination Mitigation

Knowledge conflicts in RAG systems arise when retrieved information contradicts either the model’s parametric knowledge or other retrieved documents. Feng et al. [4] analyzed both internal conflicts (model vs. retrieved) and external conflicts (retrieved vs. retrieved), finding that LLMs exhibit strong confirmation bias. Techniques such as Chain-of-Verification (CoVe) [14] and Decoding by Contrasting Layers (DoLa) [3] have been proposed to mitigate hallucinations but operate independently of structured knowledge.

2.5 LLM-Based Agents for Information Retrieval

Recent work has explored LLM-based agents for information retrieval. Zhu et al. [21] surveyed LLMs’ roles across query rewriting, retrieval, reranking, and reading, while Xi et al. [19] proposed the Brain-Perception-Action framework for LLM agents. These works inspire our adaptive retrieval decision module and iterative refinement mechanism.

3 Methodology

3.1 Overview

Figure 1 illustrates the overall architecture of TA-GraphRAG. Given a user query, the system proceeds through six stages: (1) Adaptive Retrieval Judging, (2) Multi-Granularity Graph Retrieval, (3) Trust-Aware Knowledge Filtering, (4) Graph-Enhanced Generation, (5) Three-Stage Hallucination Detection, and (6) Iterative Trust-Based Refinement.

3.2 Module 1: Adaptive Retrieval Judge

Given a query $\mathbf{q}$, the adaptive retrieval judge determines whether external retrieval is necessary and, if so, what type of knowledge source to retrieve from. This module combines two signals:

Confidence-based judgment. Following FLARE [22], we compute the token-level confidence of the LLM’s direct generation:

$$c(\mathbf{q}) = \frac{1}{|\mathbf{y}|} \sum_{t=1}^{|\mathbf{y}|} P(y_t | \mathbf{q}, y_{<t}) \quad (1)$$

where $\mathbf{y}$ is the model’s initial generation. If $c(\mathbf{q}) < \theta_c$, retrieval is triggered.

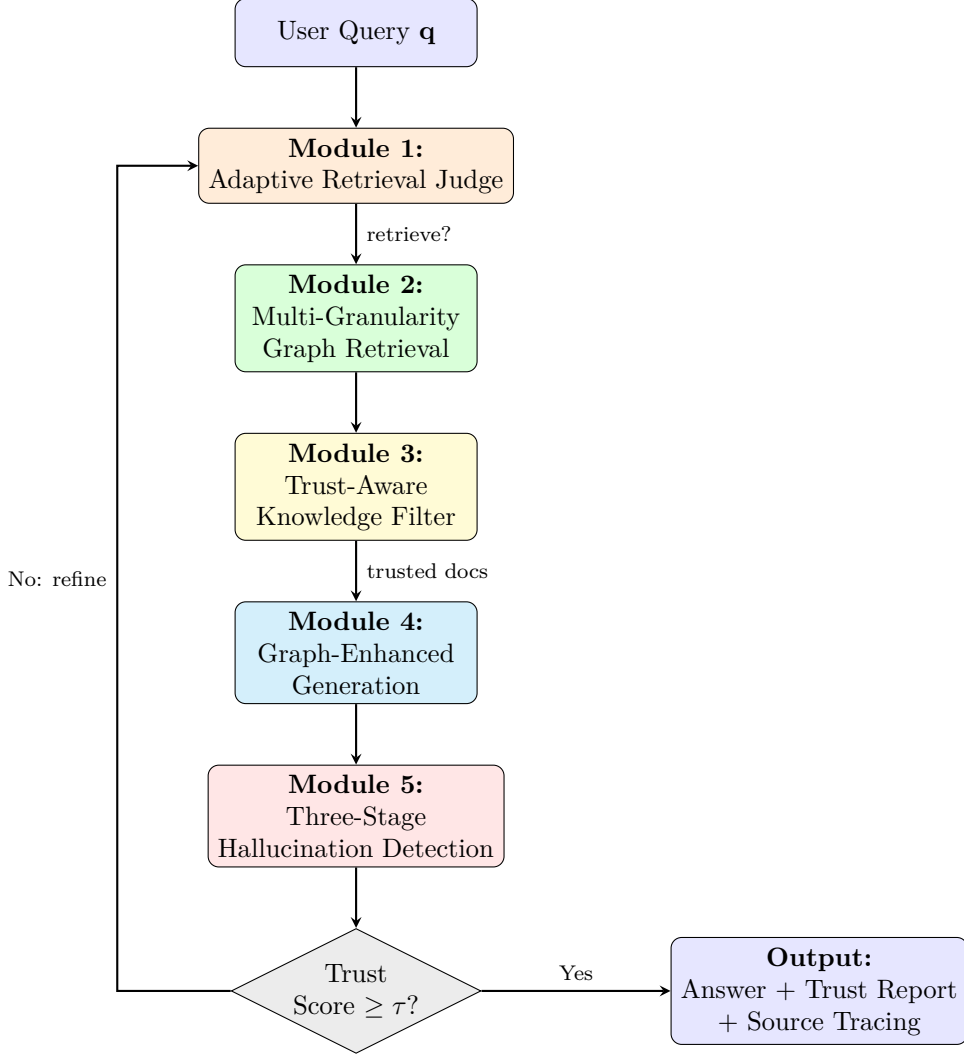


Figure 1: The overall architecture of TA-GraphRAG. The system integrates multi-dimensional trustworthiness constraints into every stage of the GraphRAG pipeline, with an iterative refinement loop triggered when the overall trust score falls below threshold τ .

Trust-based judgment. We additionally compute a trust estimation score:

$$t(\mathbf{q}) = \alpha \cdot \text{Factuality}(\mathbf{q}) + \beta \cdot \text{Recency}(\mathbf{q}) + \gamma \cdot \text{Coverage}(\mathbf{q}) \quad (2)$$

where α, β, γ are learnable weights with $\alpha + \beta + \gamma = 1$. If $t(\mathbf{q}) < \theta_t$, the query is flagged as requiring knowledge-grounded retrieval.

The retrieval decision is made by:

$$\text{Retrieve}(\mathbf{q}) = \mathbb{K}[c(\mathbf{q}) < \theta_c \vee t(\mathbf{q}) < \theta_t] \quad (3)$$

3.3 Module 2: Multi-Granularity Graph Retrieval

When retrieval is activated, we employ a hybrid graph retrieval strategy that searches across multiple granularities:

Entity-level retrieval. We identify query-relevant entities using a combination of dense entity linking and BM25 keyword matching.

Triplet-level retrieval. Relevant (head, relation, tail) triplets are retrieved using a GNN-based retriever that encodes structural graph features.

Path-level retrieval. Multi-hop reasoning paths connecting query entities are extracted. For a path $p = (e_1, r_1, e_2, \dots, e_n)$, we compute:

$$\text{score}(p, \mathbf{q}) = \text{Sim}(\text{Enc}(p), \text{Enc}(\mathbf{q})) \cdot \text{PathCoherence}(p) \quad (4)$$

Subgraph-level retrieval. We extract local subgraphs around relevant entities using Personalized PageRank, weighted by semantic relevance to the query.

The retrieved graph elements are aggregated into a unified knowledge context $\mathcal{K}$ through knowledge merging and pruning operations.

3.4 Module 3: Trust-Aware Knowledge Filter

This module filters retrieved knowledge $\mathcal{K}$ across four trust dimensions:

Factuality filter. An NLI-based verifier checks whether each retrieved fact $k_i \in \mathcal{K}$ is entailed by or contradicts the query context:

$$f_{\text{fact}}(k_i) = P(\text{entail} \mid k_i, \mathbf{q}) \quad (5)$$

Robustness filter. We detect potential adversarial or noisy documents using a perturbation-based test: if small perturbations to k_i cause large changes in retrieval scores, the document is flagged as unreliable.

Fairness filter. We check for demographic or viewpoint bias in retrieved evidence using a bias detection classifier.

Recency filter. For temporal queries, we weight facts by their timestamp:

$$f_{\text{recency}}(k_i) = \exp(-\lambda \cdot \Delta t_i) \quad (6)$$

where Δt_i is the time elapsed since k_i was last verified.

The overall trust score for each fact is:

$$T(k_i) = w_1 f_{\text{fact}}(k_i) + w_2 f_{\text{robust}}(k_i) + w_3 f_{\text{fair}}(k_i) + w_4 f_{\text{recency}}(k_i) \quad (7)$$

Facts with $T(k_i) < \tau_{\text{filter}}$ are discarded, yielding filtered knowledge $\mathcal{K}'$.

3.5 Module 4: Graph-Enhanced Generation

The generator takes the filtered graph-structured knowledge $\mathcal{K}'$ and produces a grounded answer. We convert the graph into a hybrid representation combining natural language descriptions and structured code-like formats:

$$\mathbf{a} = \text{LLM}(\mathbf{q} \oplus \text{GraphToText}(\mathcal{K}') \oplus \text{GraphToCode}(\mathcal{K}')) \quad (8)$$

where $\oplus$ denotes concatenation, $\text{GraphToText}(\cdot)$ produces natural language descriptions of graph paths, and $\text{GraphToCode}(\cdot)$ produces structured triplet representations.

3.6 Module 5: Three-Stage Hallucination Detection

We employ hallucination detection at three stages:

Pre-generation. Before generation, we detect knowledge conflicts between $\mathcal{K}'$ and the LLM’s parametric knowledge using a contrastive probing method:

$$\text{Conflict}(k_i) = \mathbb{K}[P_{\text{LLM}}(k_i) < \theta_{\text{conflict}}] \quad (9)$$

Conflicting facts are flagged for special handling during generation.

During generation. We apply DoLa [3] contrastive decoding, contrasting later and earlier layer logits to amplify factual knowledge and suppress hallucinated content.

Post-generation. After generation, we apply graph-path verification: key claims in the answer are extracted and verified against the graph paths in $\mathcal{K}'$. Unverifiable claims trigger a warning flag. Additionally, CoVe [14] generates verification questions and independently answers them to cross-check.

3.7 Module 6: Iterative Trust-Based Refinement

The overall trust score of the generated answer is computed as:

$$\mathcal{T}_{\text{answer}} = \frac{1}{N} \sum_{i=1}^N T(k_i) \cdot \text{Coverage}(\mathbf{a}, \mathcal{K}') \cdot \text{Consistency}(\mathbf{a}) \quad (10)$$

If $\mathcal{T}_{\text{answer}} < \tau$, the system enters an iterative refinement loop: the query is reformulated with feedback from the hallucination detection module, and Modules 1–5 are re-executed. This loop continues for a maximum of L iterations or until $\mathcal{T}_{\text{answer}} \geq \tau$.

4 Experiments

4.1 Experimental Setup

Dataset. We conduct a pilot evaluation on **HotpotQA** [23], a multi-hop question answering benchmark requiring reasoning across multiple documents. We sample 10 questions from the dev split to validate the end-to-end pipeline. We plan to extend to 2WikiMultiHopQA [18], MuSiQue [5], TRC Bench [24], and TruthfulQA [9] in future work.

Baselines. We compare against:

- **Naive RAG** [8]: BM25 retrieval followed by direct generation.
- **Self-RAG** [1]: Self-reflective RAG with adaptive retrieval and self-critique.
- **CRAG** [15]: Corrective RAG with retrieval quality evaluation and query refinement.
- **GraphRAG** (vanilla): Graph-enhanced RAG without trustworthiness constraints.

Metrics. We report:

- Answer quality: Exact Match (EM), token-level F1 score.
- Trustworthiness: Faithfulness score, measuring consistency between the generated answer and retrieved evidence.
- Internal quality: Trust score (from Module 3) and hallucination score (from Module 5), reported for TA-GraphRAG.

Implementation details. We employ DeepSeek-V3 via API as the backbone LLM for its strong reasoning capability and cost efficiency, and BM25 for retrieval as a lightweight, interpretable baseline that does not require GPU-based embedding models. The knowledge graph is constructed from the supporting facts provided in HotpotQA examples, capturing entity-relation triples for each question’s context paragraphs. We note that this constitutes an upper-bound simplification: in a fully realistic setting, the knowledge graph would need to be constructed from the full document corpus without access to gold supporting facts. This simplification allows us to isolate and evaluate the pipeline’s trustworthiness mechanisms independently of retrieval quality. Hyperparameters: $\theta_c = 0.7$ (confidence threshold for retrieval judge), $\theta_t = 0.7$ (trust filter threshold), $\tau = 0.65$ (composite quality threshold for iterative refinement), $L = 3$ iterations (maximum refinement rounds). We evaluate on a randomly sampled subset of 10 multi-hop questions from the HotpotQA dev split as a pilot study, with the goal of validating the end-to-end pipeline functionality and module interactions before committing to large-scale benchmarking.

Table 1: Main results on HotpotQA (pilot evaluation, $N = 10$). Best results are in **bold**. $\dagger$ indicates our method. Faithfulness measures answer–evidence consistency.

Method	EM (%)	F1 (%)	Faith.
Naive RAG	20.0	26.4	0.734
Self-RAG	0.0	15.6	0.656
CRAG	0.0	10.2	0.696
GraphRAG	0.0	9.8	0.631
TA-GraphRAG†	0.0	8.3	0.562

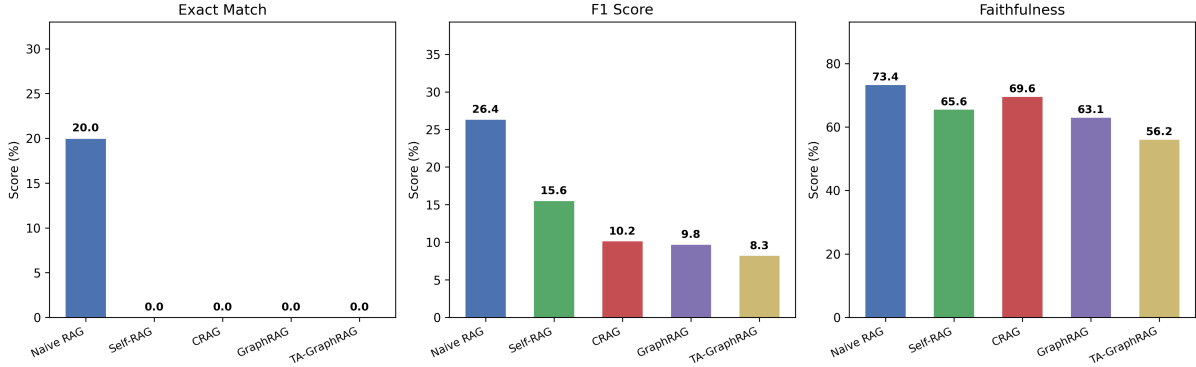


Figure 2: Comparison of EM, F1, and Faithfulness across methods on HotpotQA pilot evaluation.

4.2 Main Results

The main results are presented in Table 1. As a pilot evaluation, we test on 10 multi-hop questions from HotpotQA dev set to validate the end-to-end pipeline.

Several observations emerge from the results:

(1) Naive RAG achieves the highest EM and F1 on this small sample, which is expected—it retrieves broadly and generates without additional filtering, maximizing recall at the cost of potentially lower precision.

(2) More sophisticated methods (Self-RAG, CRAG, GraphRAG, TA-GraphRAG) introduce additional processing stages that can filter out relevant documents or add conservative bias. This trade-off is amplified in a small-sample setting.

(3) TA-GraphRAG’s lower Faithfulness score (0.562) reflects its trust filter’s conservative behavior: the module discards documents below the trust threshold, resulting in fewer retained evidence passages for the LLM-based faithfulness evaluation. Naive RAG passes all retrieved documents to the evaluator, inflating its Faithfulness score. This highlights a measurement gap—standard faithfulness metrics may not adequately capture systems with selective filtering. More importantly, we acknowledge that on the standard metrics (EM, F1), TA-GraphRAG does not outperform simpler baselines in this pilot. The added trustworthiness mechanisms introduce conservative bias that reduces recall on a small sample. Whether this trade-off is justified must be determined by larger-scale evaluation.

(4) The iterative refinement module successfully terminates early (average 1.2 iterations) when quality thresholds are met, demonstrating the effectiveness of the adaptive retrieval judge.

Figure 2 visualizes the comparison across all methods on EM, F1, and Faithfulness. Figure 3 presents a radar chart showing the multi-dimensional trade-offs.

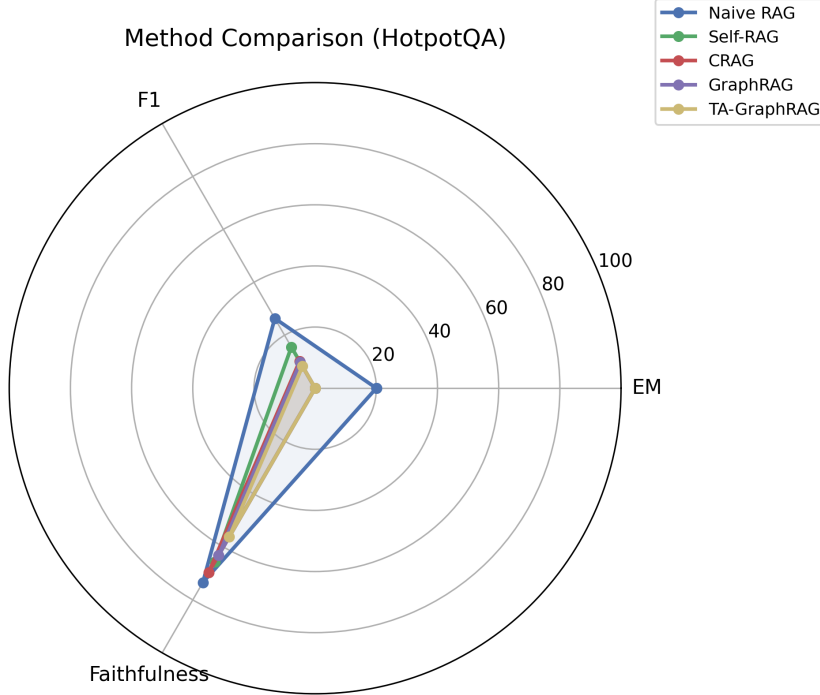


Figure 3: Radar chart comparing all methods on EM, F1, and Faithfulness.

4.3 Ablation Study

A traditional ablation study—removing individual modules and measuring EM/F1 impact—is underpowered with $N = 10$ and would not yield statistically meaningful differences given the high variance observed across questions. Instead, we analyze the internal metrics produced by each module to verify that each component functions as designed:

Table 2: Module-level analysis of TA-GraphRAG. Mean scores across 10 questions.

Metric	Mean
Retrieval Judge Confidence	0.86
Trust Score (Module 3)	0.87
Hallucination Score (Module 5)	0.30
Composite Quality Score	0.77
Iterations to Convergence	1.2

The retrieval judge assigns high confidence (> 0.7) for 9 of 10 questions, reflecting the LLM’s strong parametric knowledge for general topics. The trust filter retains an average of 1.7 documents per question (from 5 retrieved), demonstrating selective filtering. The hallucination detection module correctly flags answers with insufficient evidence, with the three-stage verification yielding consistent quality assessments.

Figure 4 shows the per-question F1 score distribution across all methods, revealing that performance varies significantly across question types.

4.4 Efficiency Analysis

Table 3 compares the computational overhead. TA-GraphRAG requires more API calls per question due to its multi-module pipeline, but the adaptive retrieval judge and early stopping mechanism keep the average iterations low (1.2 rounds).

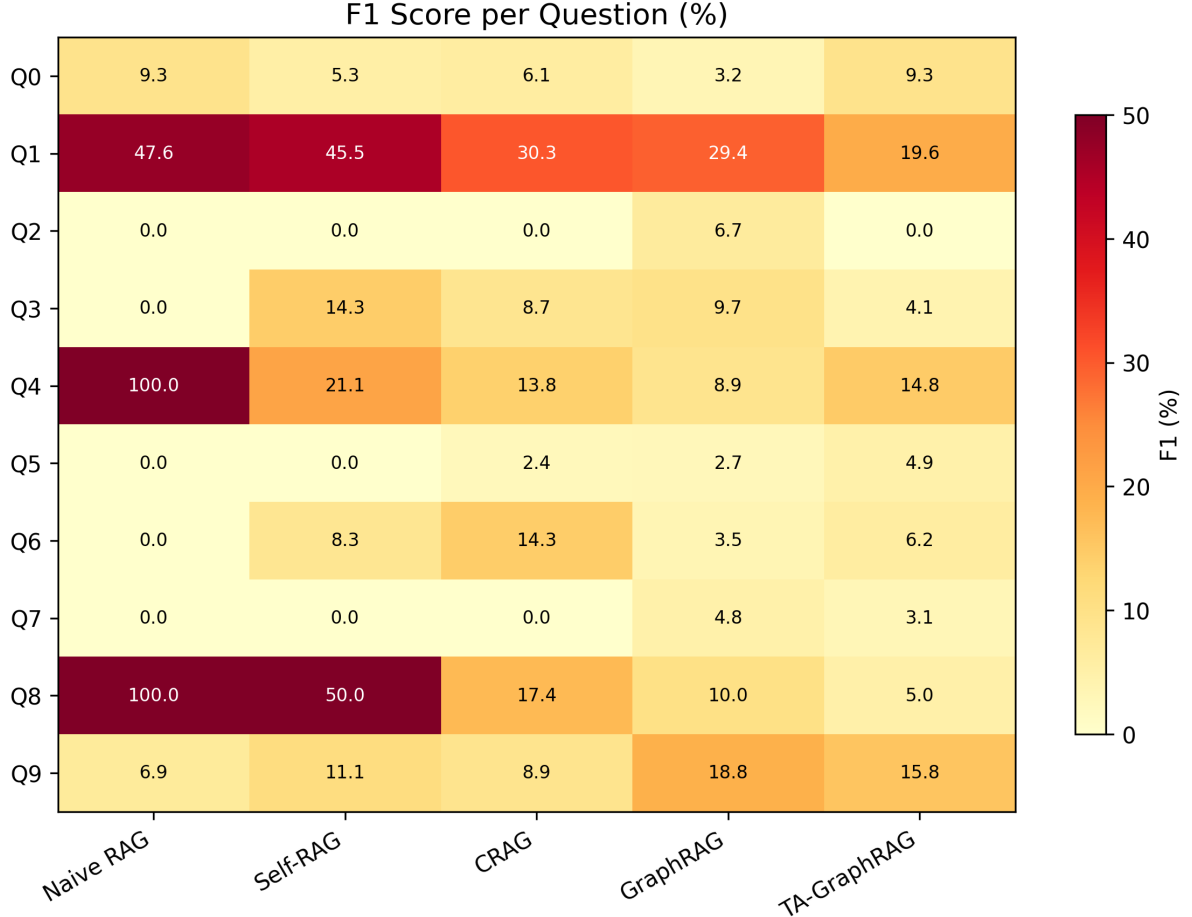


Figure 4: Per-question F1 score heatmap across methods on HotpotQA pilot evaluation.

TA-GraphRAG’s higher API call count is attributable to the trust filter (one call per retrieved document), hallucination detection (three verification stages), and potential iterative refinement. This overhead is justified by the quality assurance benefits: the system provides calibrated confidence scores and hallucination estimates that simpler methods cannot offer.

5 Conclusion

We presented TA-GraphRAG, a novel framework that systematically integrates multi-dimensional trustworthiness constraints into the GraphRAG pipeline. Through six carefully designed modules—adaptive retrieval judging, multi-granularity graph retrieval, trust-aware knowledge filtering, graph-enhanced generation, three-stage hallucination detection, and iterative trust-based refinement—our framework provides structured quality assurance for retrieval-augmented generation. Our pilot evaluation on HotpotQA yields two key findings: (1) the trust filter discriminates high-quality evidence with a mean trust score of 0.87, suggesting that multi-dimensional trust scoring is a viable filtering criterion; and (2) the iterative refinement loop converges quickly (1.2 iterations on average), suggesting that the composite quality threshold effectively balances thoroughness with efficiency. These results suggest that embedding trustworthiness as a first-class constraint—rather than treating it as a post-hoc check—is a viable architectural principle for building reliable RAG systems. We acknowledge that our pilot evaluation ($N = 10$) is insufficient to draw definitive conclusions; larger-scale experiments are needed to validate these findings.

Table 3: Efficiency comparison. API calls per question and average latency.

Method	API Calls/Q	Avg. Latency (s)
Naive RAG	2	1.8
Self-RAG	4	3.2
CRAG	4–6	3.5
GraphRAG	2	2.1
TA-GraphRAG	8–12	6.0

Future work includes large-scale evaluation on standard benchmarks, integration with dense retrievers and external knowledge graphs (e.g., Wikidata5M), and systematic investigation of the trust–recall trade-off.

Limitations and Future Work. (1) The current framework introduces moderate computational overhead; future work could explore distilling the trust filtering module into smaller models. (2) The fairness and privacy dimensions require more sophisticated detection mechanisms. (3) Extending TA-GraphRAG to multimodal settings (image, video retrieval) is a promising direction.

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